

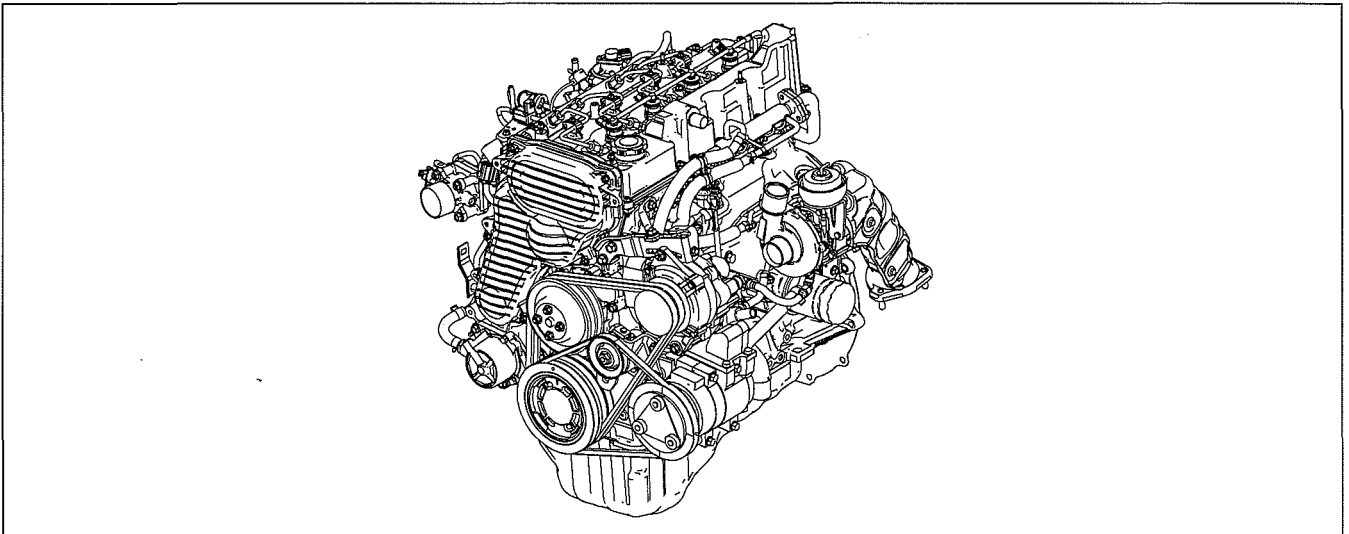
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01**ENGINE STRUCTURAL VIEW [WL-C, WE-C]**

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CYLINDER HEAD COVER CONSTRUCTION [WL-C, WE-C]

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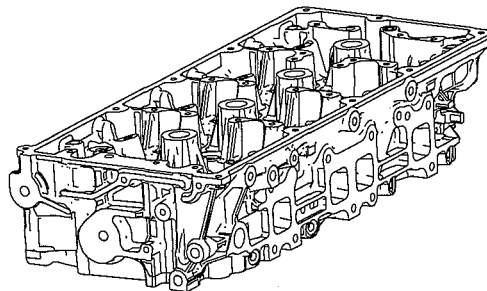
- The cylinder head cover is made of aluminum alloy.
- The cylinder head cover is made of aluminum alloy with a baffle plate on its lower side for blow-by gas separation.

MECHANICAL [WL-C, WE-C]

CYLINDER HEAD CONSTRUCTION [WL-C, WE-C]

dcf011010100t02

- A lightweight aluminum alloy cylinder head with good thermal conductivity has been adopted.
- A plastic region bolt has been adopted as a cylinder head bolt providing constant high axial force (tightening force).
- Due to the adoption of 4 valves, the opening area has been increased and the charging efficiency improved.
- The smooth curved intake port reduces intake resistance.



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CYLINDER HEAD GASKET CONSTRUCTION [WL-C, WE-C]

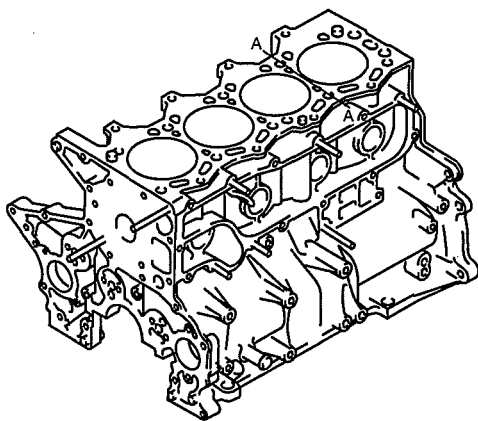
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- A steel-laminated gasket has been adopted for the cylinder head gasket.

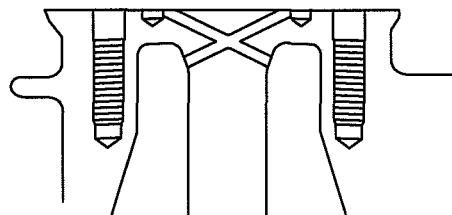
CYLINDER BLOCK CONSTRUCTION [WL-C, WE-C]

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- A special cast iron liner-less type cylinder block has been adopted to provide a thin-walled cylinder block. In addition, a curve-shaped, deep-skirt cylinder block has been adopted to heighten the cylinder block rigidity.
- The engine coolant passage has been equipped by cross-drilling in the bore, improving the cooling performance and reducing oil consumption.



SECTION A—A
CROSS DRILL HOLE



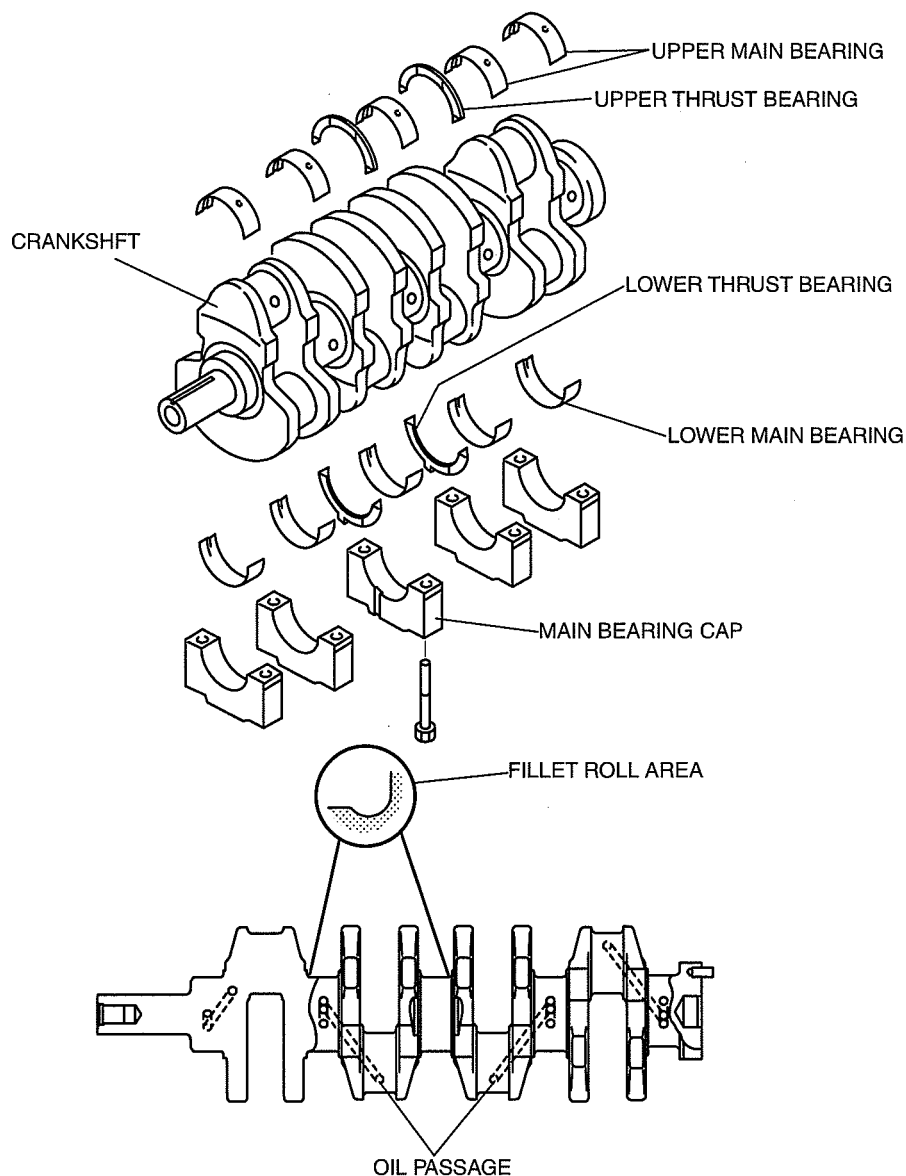
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CRANKSHAFT, MAIN BEARING CONSTRUCTION [WL-C, WE-C]

dcf011011301102

Structure

- A five axle-hole, 8 counter weight steel crankshaft has been adopted.
- An oil line for supplying oil to each journal is provided in the crankshaft. Crank pins and fillets on both sides of the journal are rolled to bear heavy loads.
- Upper and lower main bearings are made of aluminum alloy and thrust bearings are used at the No.3 journal. The upper main bearing has oil grooves and oil holes.



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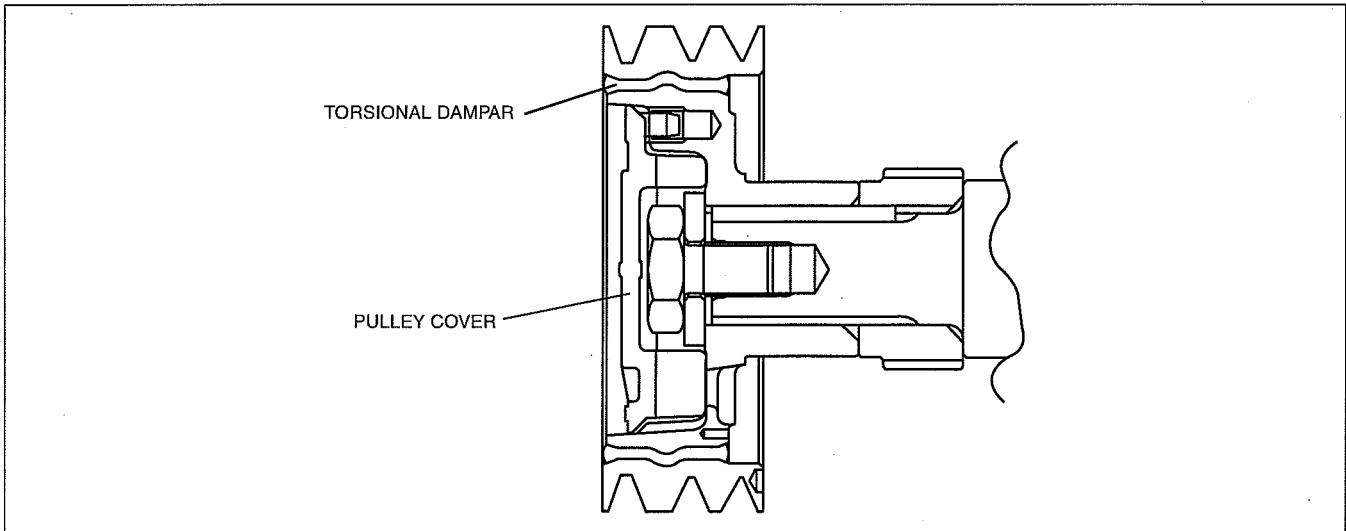
MECHANICAL [WL-C, WE-C]

CRANKSHAFT PULLEY CONSTRUCTION [WL-C, WE-C]

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Structure

- A torsional damper pulley is used for the crankshaft pulley to reduce torsional vibration noise.
- The pulley installation area has a pulley cover for reduction of engine noise.



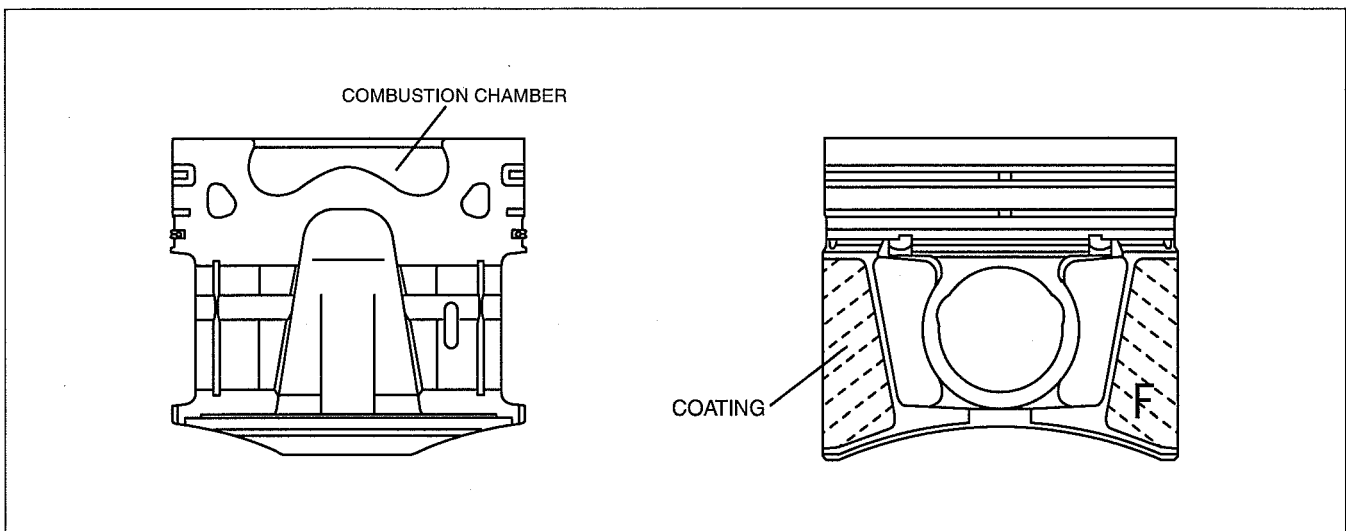
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PISTON CONSTRUCTION [WL-C, WE-C]

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Structure

- In accordance with the adoption of the common rail injection system, the piston head shape (combustion chamber) has been changed.
- A reduction in friction has been achieved by the coating of the skirt area together with a compression rating setting of 18.0.
- To realize high combustion efficiency, the diffusion area of the hotspot at combustion has been expanded by the change in the shape of the combustion chamber and the spray pattern of fuel injectors. Exhaust emission and combustion noise have been reduced through smooth and uniform combustion.



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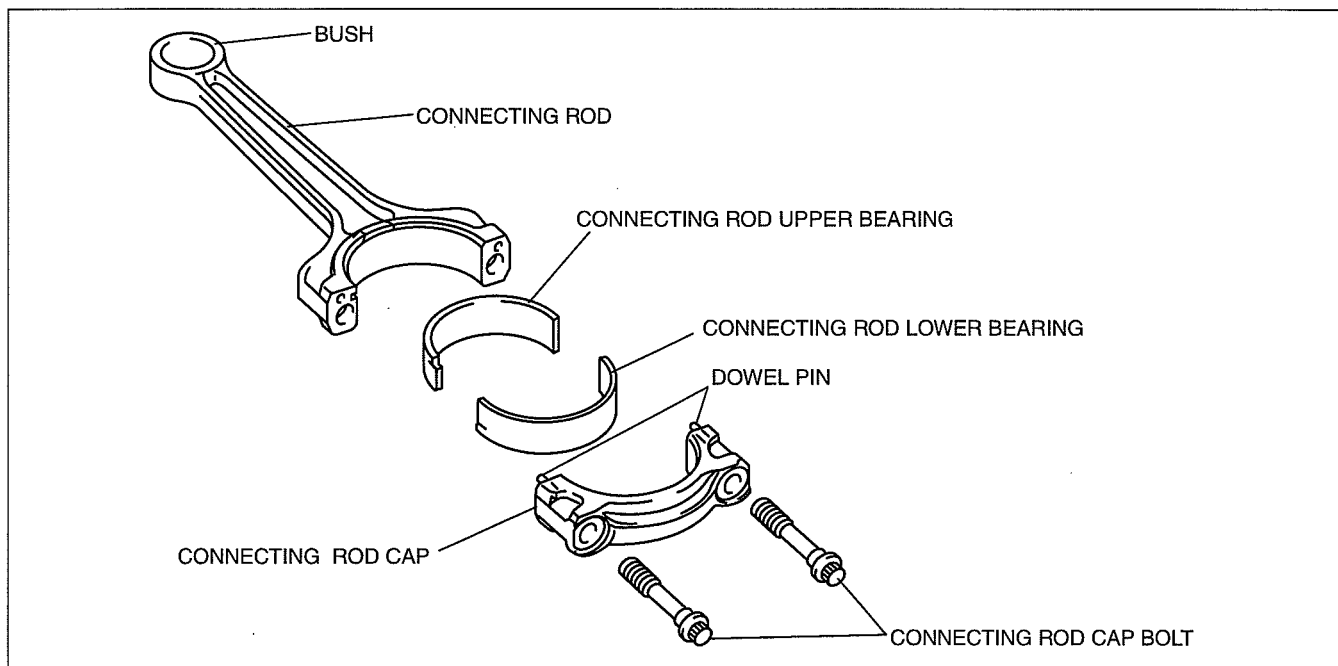
MECHANICAL [WL-C, WE-C]

CONNECTING ROD, CONNECTING ROD BEARING CONSTRUCTION [WL-C, WE-C]

dcf01101121102

Structure

- Connecting rods are made of forged manganese steel.
- Dowel pins are used to position the rod cap to the connecting rod.
- Connecting rod caps are secured with plastic region bolts.
- The connecting rod bearing is made of aluminum alloy.



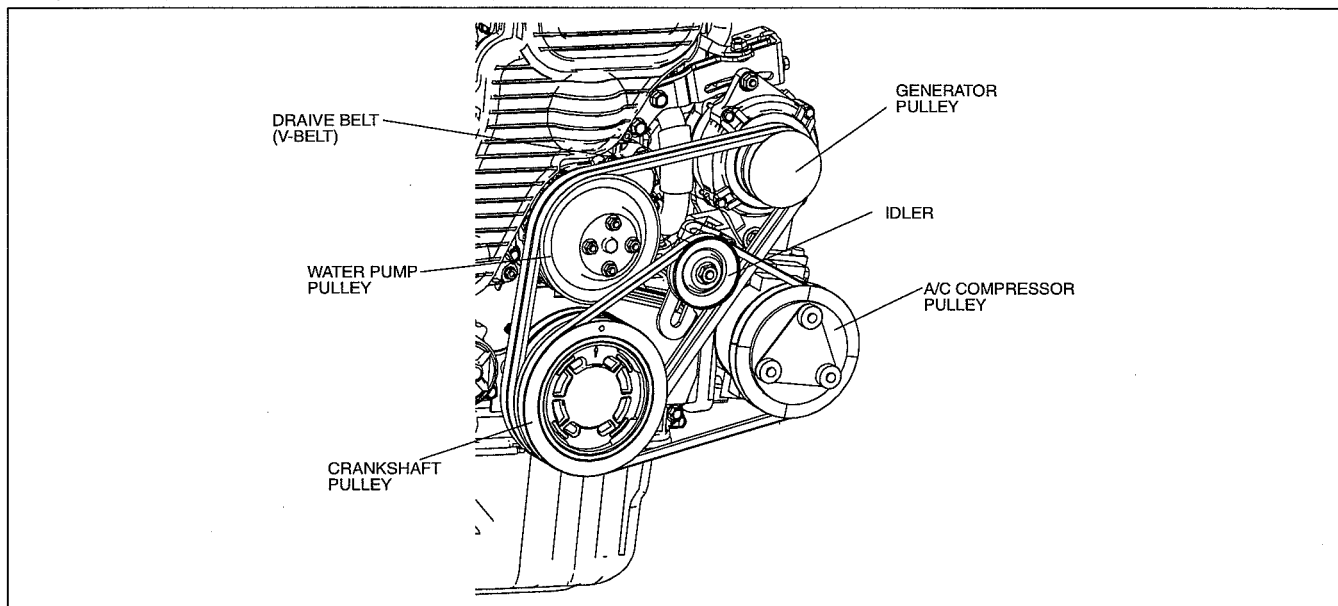
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DRIVE BELT CONSTRUCTION [WL-C, WE-C]

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Structure

- A V-belt has been adopted which consists of three belts including generator and water pump operation belts (two belts), and an A/C compressor operation belt.

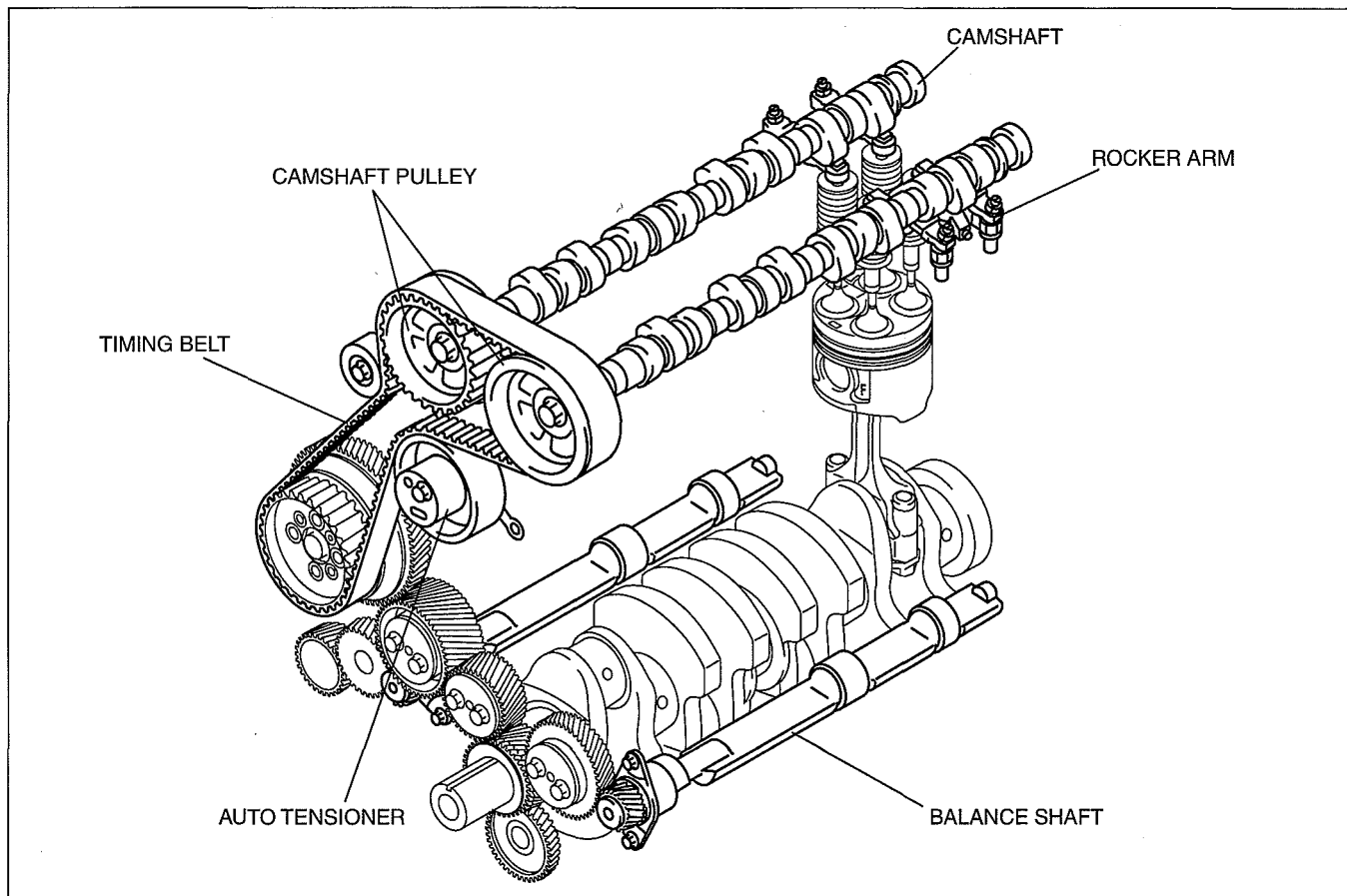


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MECHANICAL [WL-C, WE-C]

VALVE MECHANISM STRUCTURAL VIEW [WL-C, WE-C]

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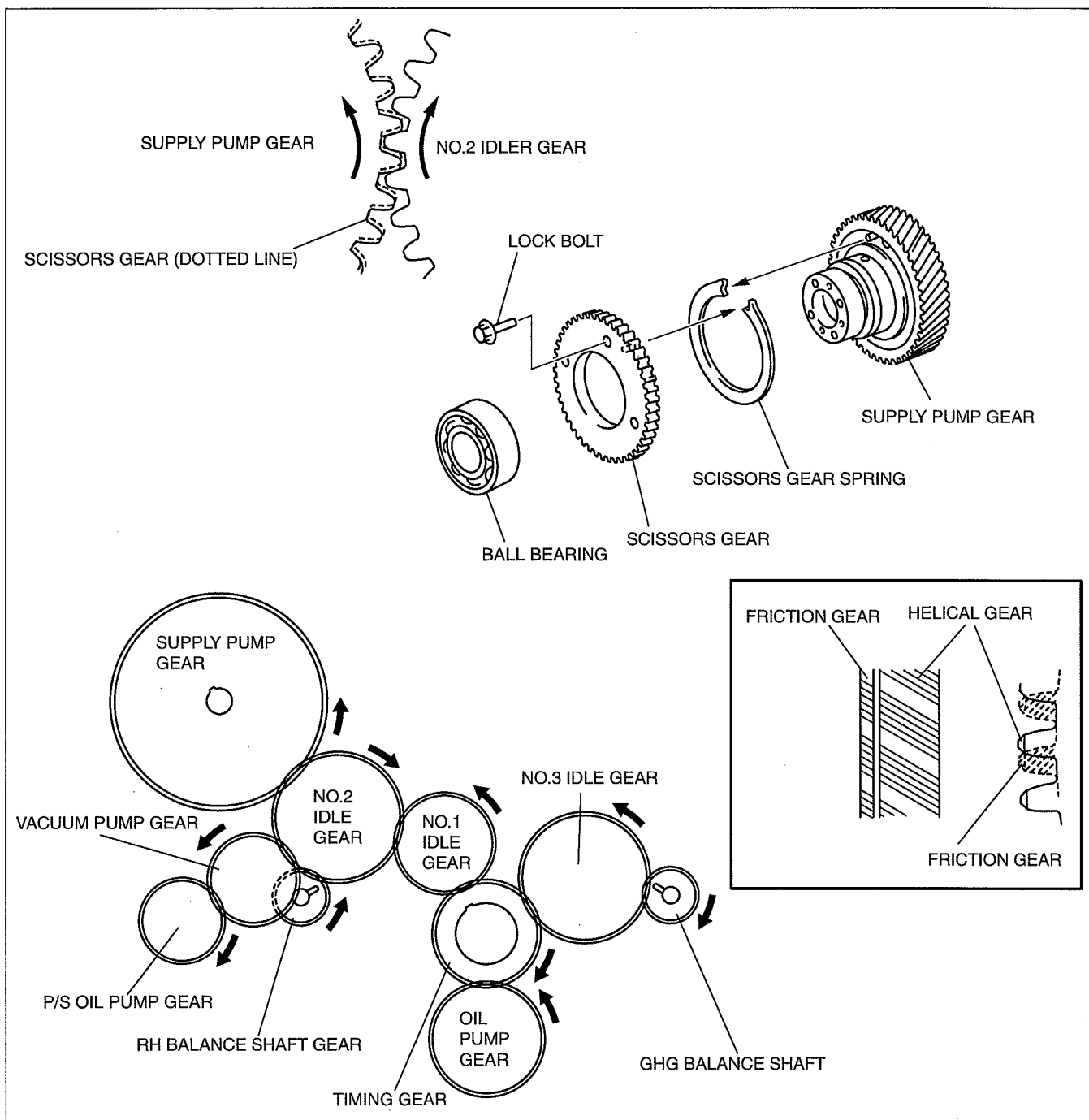
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TIMING GEAR CONSTRUCTION [WL-C, WE-C]

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Supply Pump Gear

- The gears are helical. Supply pump gear is combined with a scissors gear.
- The scissors gear, through the force of the scissors gear spring, prevents the teeth of the supply pump gear from moving back and forth (backlash) between the teeth of the No.2 idler gear, with which the supply pump gear is meshed, and thus reduces gear noise.
- When removing the supply pump gear, fix the scissors gear to the supply pump gear using a lock bolt (M6 x 1.0, length under the bolt head is approximately 10 mm {0.63 in}) to prevent the scissors gear from rotating under the spring force.



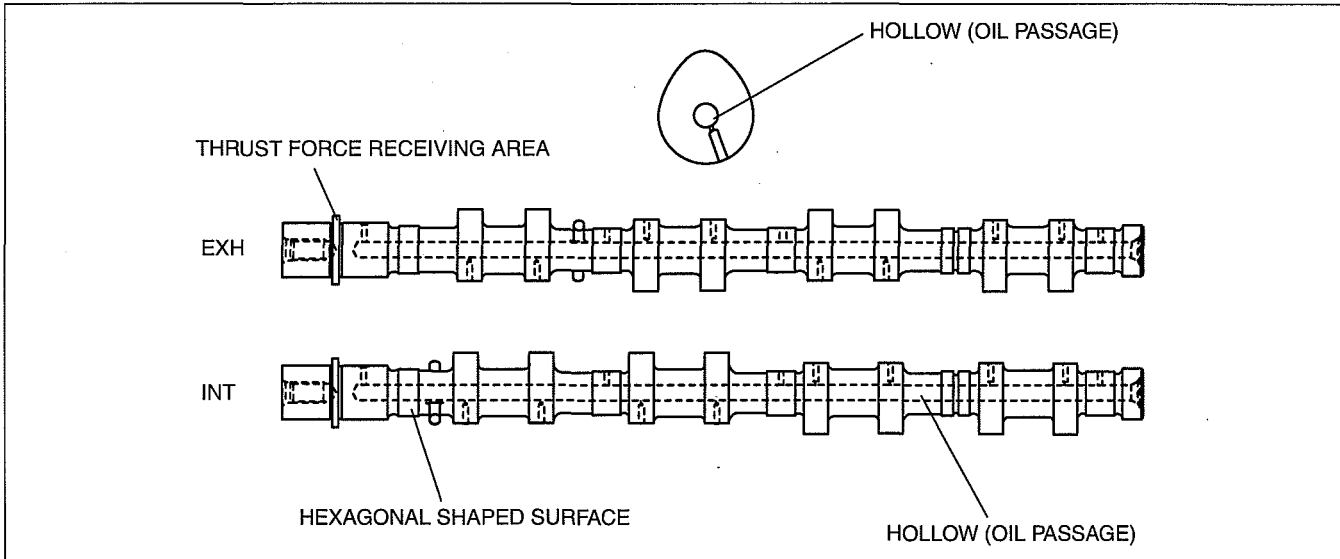
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MECHANICAL [WL-C, WE-C]

CAMSHAFT CONSTRUCTION [WL-C, WE-C]

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- The camshaft is made of cast iron alloy.
- A hollow type camshaft with five bearings has been adopted. In addition, the hollow area is provided as the oil passage.
- The camshaft No. 1 journal has a hexagonal shaped surface around the rear of the shaft to prevent rotation during servicing.
- Thrust force is received at the front of the No. 1 journal.

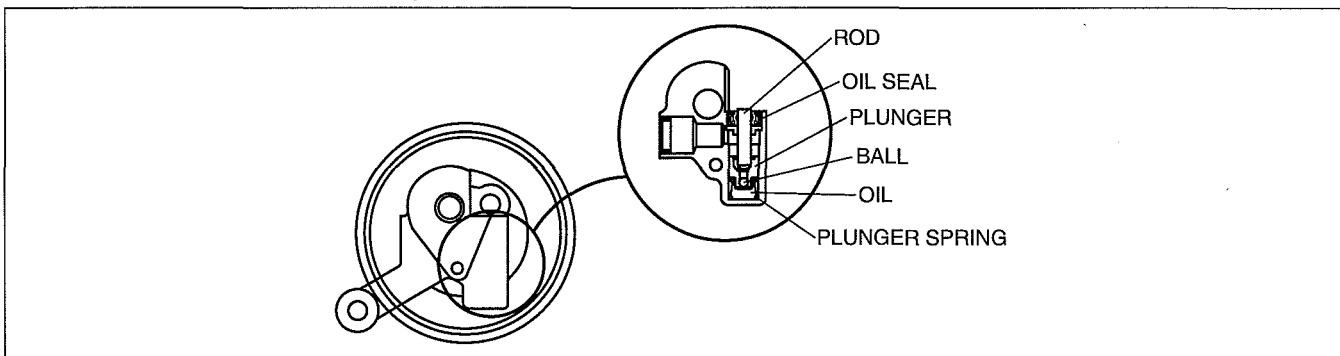


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AUTO TENSIONER CONSTRUCTION [WL-C, WE-C]

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- A hydraulic timing belt auto tensioner has been adopted for timing belt adjustment so that the timing belt tension is maintained constantly.



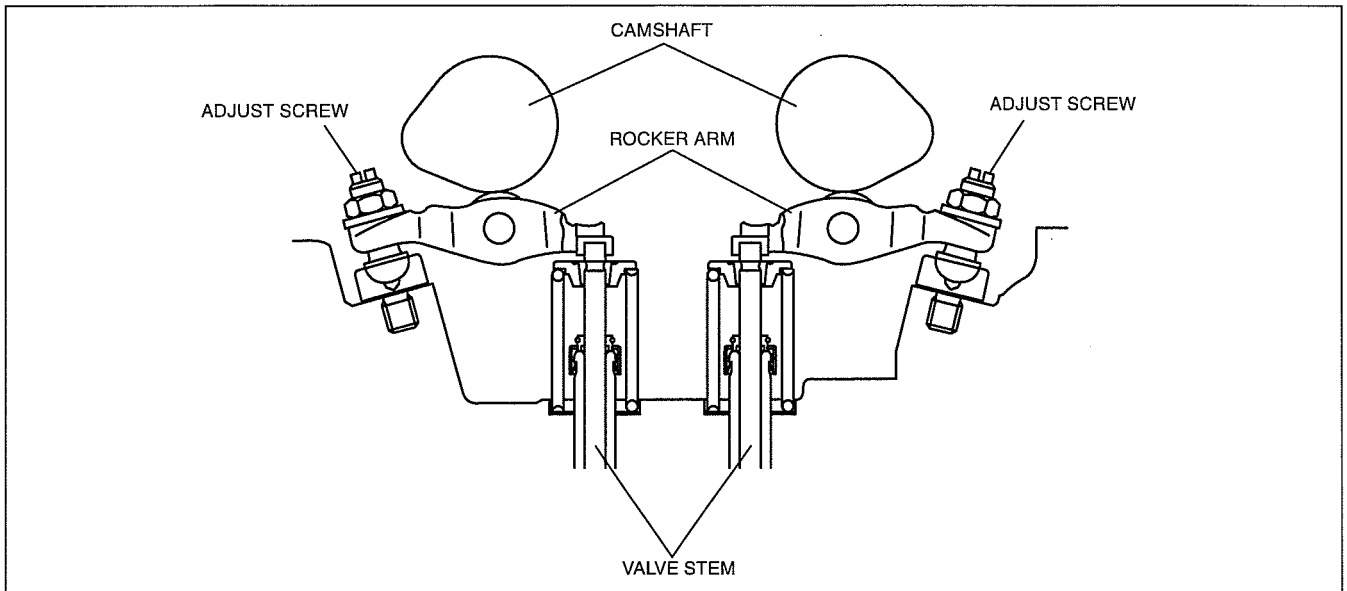
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MECHANICAL [WL-C, WE-C]

VALVE, ROCKER ARM CONSTRUCTION [WL-C, WE-C]

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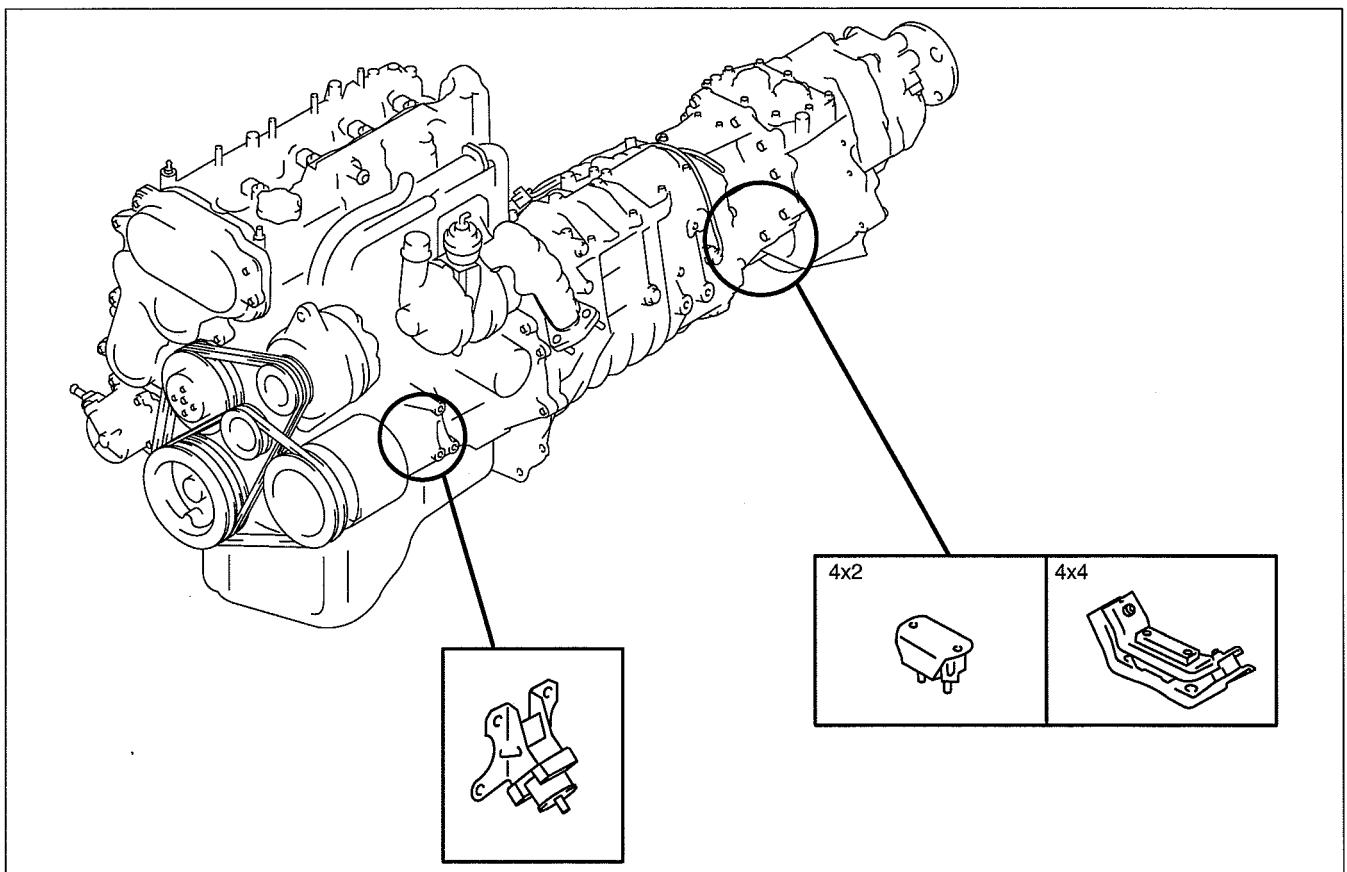
- The valves are operated by rocker arms.
- Valve clearance is adjusted by the adjust screw and nut of the rocker arm.



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ENGINE MOUNT STRUCTURAL VIEW [WL-C, WE-C]

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DBG110BTB809

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